ETL Pipeline Exercise: Transform

Explore how to transform extracted data within an ETL pipeline by reformatting date fields from numeric to text, cleaning text columns of tabs and new lines, and categorizing follower counts into defined levels. Understand how to implement these transformations using Python functions integrated with Apache Airflow to prepare data for loading into warehouses. This lesson helps you refine and structure data to meet business and schema requirements, and export the results for subsequent use.

To continue our pipeline implementation, we'll now focus on transforming the extracted data. According to the business requirements and the schema of the data warehouse, there are a few issues we need to fix with our extracted data. They are:

1. To change the month format of all date columns from numerical to text (for example, from `08` to `Aug`)
2. To remove tabs and new lines from columns `comment_text` and `post_text`
3. To bin the number of followers into three categories, `low`, `medium`, `high` (the number of followers lower than 1000 will be in `low` category, the number of followers between 1000 and 5000 will be in `medium` category, and higher than 5000 will be in `high` category)

As before, we'll write a function that accomplishes these tasks in a file called `helper.py`. Then, we'll import the function to the DAG to add it to our pipeline.

> **Note**: Fill in the missing code snippets.

Ask

Change the month format

As mentioned, the first task in the transform stage is to change the month format of columns: `post_date` and `comment_date` from numerical to text. Use the command `strftime` to modify the format of datetime objects.

helper.py Python

```python
import pandas as pd

def transform():
    df = pd.read_csv('/airflow/dags/social_media_data.csv')

    # Modify the post_date and comment_date columns to be datetime object
    df["post_date"] = pd.to_datetime(df.post_date)
    df["comment_date"] = pd.to_datetime(df.comment_date)

    # change the format to YYYY-MM-DD HH:MM:SS
    df["post_date"] = df[<<InsertHere>>].dt.<<InsertHere>>('%Y-%b-%d %H:%
    df["comment_date"] = df[<<InsertHere>>].dt.<<InsertHere>>('%Y-%b-%d %
```

Run

Changing the month format

Remove tabs and new lines

The next task in the transform stage is to remove all tabs (`'\t'`) and new lines (`'\n'`) from the text columns `post_text` and `comment_text`. Use the replace method to find a character in a string and replace it with another.

helper.py Python

Ask

```python
import pandas as pd

def transform():
    df = pd.read_csv('/airflow/dags/social_media_data.csv')
```

```
    # Modify the post_date and comment_date columns to be datetime object
    df["post_date"] = pd.to_datetime(df.post_date)
    df["comment_date"] = pd.to_datetime(df.comment_date)

    # change the format to YYYY-MM-DD HH:MM:SS
    df["post_date"] = df["post_date"].dt.strftime('%Y-%b-%d %H:%M:%S')
    df["comment_date"] = df["comment_date"].dt.strftime('%Y-%b-%d %H:%M:%
```

Run

Removing the tabs and new lines

Bin `followers`

The final task in the transform stage is to create a new column called `followers_level`, which categorizes the number of followers into three categories: `low`, `medium`, and `high`.

- `followers` lower than 1000 will have a `followers_level` value of `low`.
- `followers` between 1000 and 5000 will have a `followers_level` value of `medium`.
- `followers` higher than 5000 will have a `followers_level` value of `high`.

Finally, export the transformed DataFrame as a CSV file called `transformed_social_media_data.csv`. At the end of the script, print a sample of the transformed data.

helper.py Python

```
import pandas as pd

def transform():
    df = pd.read_csv('/airflow/dags/social_media_data.csv')

    # Modify the post_date and comment_date columns to be datetime object
    df["post_date"] = pd.to_datetime(df.post_date)
    df["comment_date"] = pd.to_datetime(df.comment_date)
```

Ask

```
    # change the format to YYYY-MM-DD HH:MM:SS
    df["post_date"] = df["post_date"].dt.strftime('%Y-%b-%d %H:%M:%S')
    df["comment_date"] = df["comment_date"].dt.strftime('%Y-%b-%d %H:%M:%
```

Run

Binning the column followers

Here are the solution codes for the above tasks:

Solution code: Change the month format

helper.py Python

```
import pandas as pd

def transform():
    df = pd.read_csv('/airflow/dags/social_media_data.csv')

    # Modify the post_date and comment_date columns to be datetime object
    df["post_date"] = pd.to_datetime(df.post_date)
    df["comment_date"] = pd.to_datetime(df.comment_date)

    # change the format to YYYY-MM-DD HH:MM:SS
    df["post_date"] = df["post_date"].dt.strftime('%Y-%b-%d %H:%M:%S')
    df["comment_date"] = df["comment_date"].dt.strftime('%Y-%b-%d %H:%M:%
```

Run

Solution code: Change the month format

- **Line 4**: We load the extracted data from `social_media_data.csv`.
- **Lines 7–8**: We convert the data type of the columns `post_date` and `comment_date` to datetime.

Solution code: Remove tabs and new lines

Ask

helper.py Python

```
import pandas as pd

def transform():
    df = pd.read_csv('/airflow/dags/social_media_data.csv')

    # Modify the post_date and comment_date columns to be datetime object
    df["post_date"] = pd.to_datetime(df.post_date)
    df["comment_date"] = pd.to_datetime(df.comment_date)

    # change the format to YYYY-MM-DD HH:MM:SS
    df["post_date"] = df["post_date"].dt.strftime('%Y-%b-%d %H:%M:%S')
    df["comment_date"] = df["comment_date"].dt.strftime('%Y-%b-%d %H:%M:%
```

Run

Solution code: Remove tabs and new lines

- **Lines 15–16**: We remove all tabs and new lines from columns `comment_text` and `post_text`.

Solution code: Bin `followers`

helper.py Python

```
import pandas as pd

def transform():
    df = pd.read_csv('/airflow/dags/social_media_data.csv')

    # Modify the post_date and comment_date columns to be datetime object
    df["post_date"] = pd.to_datetime(df.post_date)
    df["comment_date"] = pd.to_datetime(df.comment_date)

    # change the format to YYYY-MM-DD HH:MM:SS
    df["post_date"] = df["post_date"].dt.strftime('%Y-%b-%d %H:%M:%S')
    df["comment_date"] = df["comment_date"].dt.strftime('%Y-%b-%d %H:%M:%
```

Ask

Solution code: Binning followers

- **Lines 19–22**: We bin the `followers` column into the categories `low`, `medium`, and `high`.
- **Line 25**: We export the transformed DataFrame as a CSV file named `transformed_social_media_data.csv`.

Ask